Identification of Jets at the LHC using Raw Calorimeter Images - Miniproject Report

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Abstract

This study presents a machine-learning analysis of five-class jet classification at the Large Hadron Collider (LHC) using simulated calorimeter images of jets intiated by gluons, quarks, W bosons, Z bosons and Top quarks. Two classification approaches are compared: a convolutional neural network (CNN) trained on 100x100 pixel calorimeter images, and a dense neural network (DNN) trained on 53 pre-selected jet substructure features with associated particle identifications. The image CNN achieves a validation accuracy of 72% and a macro-average AUC of 0.926, demonstrating physically meaningful jet substructure can be extracted on pure calorimeter image data without any explicit physics input. Moreover, analysis of per-class classification efficiency as a function of jet mass reveals that the CNN implicitly recovers the characteristic mass peaks of the W boson at approximately 80GeV and Z boson at approximately 91 GeV which is consistent with known masses, despite receiving no explicit mass information during training. However, the CNN struggles to separate W and Z jets at high transverse momentum, suggesting the mass difference between two bosons becomes unresolvable from pixel patterns alone at high transverse momenta, which is reflected in a Z-W misclassification rate of 21%. The DNN, trained only on jet features, outperforms the image CNN, with a validation accuracy of 81% and a macro-averaged AUC of 0.962. The most striking improvement is the fivefold reduction in Z-W misclassification from 21% to 4%. These results demonstrate that while CNN-based jet image classifiers can learn physical substructure from raw calorimeter data, explicit jet substructure observables provide more discriminating power for identifying kinematically similar particles.

1 Background + Introduction

Particle jets are narrow collimated sprays of hadrons produced in high-energy proton-proton collisions, which cause high-energy quarks or gluons to undergo fragmentation and hadronisation at the Large Hadron Collider (LHC) (Salam, G.P. 2010). The identification and classification of jets is one of the most important tasks in experimental particle physics as jet substructure has provided numerous innovative new ways to search for new physics and to probe the Standard Model in extreme regions of phase space (Larkoski et al. 2020). Jets can originate from different particles, which each have a distinct substructure that reflect the kinematics of the originating particle and its decay products (Abdesselam, A. et al. 2011, Larkoski et al. 2020).

Traditionally, the approaches used for jet classification rely on observables of the jets, designed by physicists, which include variables such as jet mass, N-subjettiness, and jet multiplicity, which can capture discriminating features of jet substructure (Thaler, J. and Van Tilburg, K. 2011). While powerful, these approaches require substantial expertise in QCD theory and detector physics. This reality is demonstrated by Adams et al. (2015) with the BOOST2013 workshop report, which reviews the entire jet substructure observables proposed throughout decades of literature. Furthermore, jet classification reliant upon jet feature observables may not capture all available information in the raw detector data, since compressing a full calorimeter image into a reduced set of scalar observables inevitably discards information (Grojean & Moore. 2024).

Thus, the advent of deep learning has presented an alternative approach for jet identification, by translating the calorimeter energy deposits of a jet into a two-dimensional image and applying convolutional neural networks (CNNs) directly to the pixel-level data. This new paradigm was established by de Olivereira et al. (2016) who demonstrated that CNNs trained on jet images can match feature-based classifiers. Nonetheless, the image-based CNN approach has notable limitations. Due to the sparse nature of jet images, typically 90% of pixels are empty, leading to greater difficulty for convolutional filters to capture key substructure properties efficiently. In conjunction, each pixel only contains information about where and how much energy is deposited; there is no other discriminatory information that can be used for a simple image-based CNN classification (Qu & Gouskos 2020).

This project applies both approaches to the five-class jet classification problem using the publicly available HLS4ML LHC jet dataset (Pierini, M. et al. 2020), which comprises of 100x100 pixel calorimeter images of jets from five types of particles: gluons, light quarks, W bosons, Z bosons, and top quarks. The three main tasks set out in this project are: to build, train and develop a CNN classifier that can separate the five jet classes using only the calorimeter image data; investigate how classification efficiency depends on jet feature variables, namely transverse momentum, jet mass and constituent multiplicity; and compare model performance with a dense neural network (DNN) trained on 53 jet substructure observables to quantify how much discriminating information is contained in the raw pixel representation versus observed substructure features.

2 Dataset and Preprocessing

2.1 Dataset

For this study, we use the HLS4ML LHC jet datset, wich consists of simulated proton-proton collisions collisoin events generated using the PYTHIA 8 Monte Carlo generator (Sjöstrand et al. 2015), which models the hard scattering process, parton showering and hadronisation. Detector effects are simulated using the DELPHES fast simulation framework (de Favereau et al. 2014), producing the calorimeter energy deposites that form the jet images. Jets are reconstructed using the anti-kT clustering algorithm (Cacciari et al. 2008) as implemented in FastJet (Cacciari et al. 2012). This procedure follows the same outline as described in Coleman et al. (2018), which is the study from which the HLS4ML dataset is derived.

In this dataset, each jet is represented as 100x100 calorimeter images, labelled according to its originating particle of five mutually exclusive classes: gluon, light quark, W boson, Z boson, and top quark. Three image types are available per jet: the combined calorimeter image (jetImage), the electromagnetic calorimeter image (jetImageECAL) and the hadronic calorimeter image (JetImageHCAL). This study only uses the combined jetImage as input to the CNN classifier.

Alongside the calorimeter images, the dataset provides 53 jet-level variables per event, including kinematic variables, jet substructure observables, and constituent multiplicity.

2.2 Train/validation split

The dataset files are indexed by a file number prefix. Files with index 0-6 are used for training files and files with index 7-9 are used for validation. Due to the structure of the publicly available dataset, no files specifically held out as test files are provided, thus, the effective training set used was an 80% split of the readily available training set, with the remainder constituting the test set. Due to dataset size constraints imposed by available RAM, a maximum of 8 training files and 3 validation files were loaded, yielding 80,000 training jets (split into 64,000 jets for training and 16,000 jets for testing) and 30,000 validation jets respectively. As the validation set was only used for early stopping monitoring and not used to update model weights directly, the validation set is also used a reasonable proxy for unseen data, though this limitation is acknowledged in this study.

2.3 Log-normalisation preprocessing

The raw pixel values in the jet images span several orders of magnitude. Thus, to map these values into a well-conditioned range suitable for neural network training, a log-normalisation transform is applied to all images, given by:

$$I_{norm} = \frac{\ln(clip(I, v_{min}, v_{max})) - \ln(v_{min})}{\ln(v_{max}) - \ln(v_{min})} \tag{1}$$

where $v_{min} = 0.01$ and $v_{max} = 10^3$ are chosen and the clip operation floors all pixel values to v_{min} before taking the logarithm to avoid undefined values at zero. This normalisation ensures values lie in the range [0,1], which is optimal for training the CNN model.

For the features DNN, the 53 jet substructure observables are standardised using sci-kit-learn's StandardScaler (Pedregosa, F. et al. 2011), which subtracts the training set mean and divides by training set deviation for each feature independently. This process is necessary as feature variables span very different numerical ranges (ie. transverse momentum ranges from 400 to 2500GeV while dimensionless ratios range from 0 to 1). The scalar is fitted on the training set only and applied without re-fitting on the validation set, such that the validation does not influence the training process.

2.4 Data shuffling

After preprocessing, both image and feature arrays are shuffled using a fixed random seed of 42 via sci-kit-learn's shuffle array function (Pedregosa, F. et al. 2011). A fixed seed ensure results are reproducible across runs. Shuffling is necessary as the data is loaded sequentially. Without shuffling, each mini-batch would contain jets predominantly from one file and potentially one class region, which may cause unintended biasing in gradient estimates and lead to poor convergence.

3 Model Architectures

3.1 Image Convolutional Neural Network (CNN) Model

The image classifier is a CNN that takes a single 100x100 pixel calorimeter image as input and outputs a probability distribution over the five jet classes. The architecture consists of three convolutional blocks followed by a fully connected classification head. Each block applies a Conv2D layer with a 3x3 kernel and same padding, and a 2x2 Maxpooling window, which follows the approach established by Simonyan, K. and Zisserman, A. (2015) who demonstrated that stacking multiple 3x3 layers achieves an equivalent receptive field to a single larger kernel with fewer parameters and greater non-linearity. However, it is worth noting that de Oliveira et al. (2016) found that an 11x11 initial kernel gave a higher AUC in their jet image CNN, suggesting that for jet images specifically larger initial kernels may be moer appropriate for capturing global substructure patterns. The 3x3 choice in this work prioritises parameter efficiency and training stability over maximising receptive field in the first layer, a deliberate trade-off to allow the model to run in a reasonable timeframe given the GPU RAM constraint present in this study.

The LeakyReLU activation function is used in each convolutional block instead of the standard ReLU activation. ReLU activation may cause dead neuron problems since most of the image pixel values are zero for sparse inputs, causing pre-activation inputs to be negative, thus causing the gradient to become zero. LeakyReLU activation function is given by,

$$\text{LeakyReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha x & \text{if } x \leq 0 \end{cases} \tag{2}$$

where $\alpha = 0.1$ is the negative slope coefficient. This activation addresses this gradient problem by outputting a small negative value for negative inputs rather than exactly zero, maintaining a gradient flow through all neurons even on predominantly-zero jet images (Maas, A.L. et al. 2013). This is essential for training stability on the HLS4ML jet image dataset, where approximately 96% of pixels have a value of 'zero'.

L2 weight regularisation with penalty coefficient 1×10^{-4} is applied to all convolutional and dense layers to penalise large weights and reduce overfitting. Dropout with drop probability 0.3 is applied after the Dense layer, which randomly deactivates 30% of neurons during each training step. This prevents the neurons from co-adapting too much, which consequently further reduces over-fitting as well as improves performance of the neural network (Srivastava, N. et al. 2014).

The model is compiled with the AdamW optimiser (Loshchilov, I. et al. 2019) with learning rate 1×10^{-3}, SparseCategroicalCrossentropy loss, and early stopping with patience 10. A summary of the CNN architecture is given in Table 1.

Table 1: Image CNN (model_v1) architecture summary

Layer	**Output Shape**	**Parameters**
Conv2D (32 filters, 3×3) + LeakyReLU + MaxPool	$50 \times 50 \times 32$	320
Conv2D (64 filters, 3×3) + LeakyReLU + MaxPool	$25 \times 25 \times 64$	18,496
Conv2D (128 filters, 3×3) + LeakyReLU + MaxPool	$12 \times 12 \times 128$	73,856
Flatten	18,432	0
Dense (256) + LeakyReLU + Dropout (0.3)	256	4,718,848
Dense (5) — output	5	1,285
Total		**4,812,805**

3.2 Features Dense Neural Network (DNN) Model

The features classifier is a fully connected DNN that takes 53 standardised jet substructure observables as a flat input vector and output a probability distribution over the five jet classes. As this model is only trained on jet features, it has no spatial structure since jet feature variables have no meaningful spatial relationship to each other.

The architecture consists of three dense blocks, each containing a Dense layer, BatchNormalization (Ioffe, S. and Szegedy, C. 2015), LeakyReLU activation and Dropout with drop probability of 0.3, using 256, 256 and 128 neurons in each respective layer. The decreasing layer widths act to compress the 53-dimensional feature input representation toward the 5-dimensional output space. The relatively wide first two layers are chosen to give the DNN a comparable capacity to the image CNN, ensuring that any performance difference between the two models reflects the input representation rather than a disparity in model complexity.

The same training configuration as for the CNN is thus used, yielding the following architecture summary in Table 2:

Table 2: Features DNN architecture summary

Layer	**Output Shape**	**Parameters**
Dense (256) + BatchNorm + LeakyReLU + Dropout (0.3)	256	13,312
Dense (256) + BatchNorm + LeakyReLU + Dropout (0.3)	256	66,816
Dense (128) + BatchNorm + LeakyReLU + Dropout (0.3)	128	33,408
Dense (5) — output	5	645
Total trainable		**115,717**

4 Results - Image CNN

4.1 Training Performance

The image CNN was trained on 64,000 jets from HLS4ML dataset using the Adam optimiser with early stopping monitoring validation loss. Training converged after 18 epochs, at which point early stopping retained the weights from the epoch with the lowest validation loss. The training and validation accuracy curves (left plot) are shown with corresponding loss curves (right plot) in Figure 1.

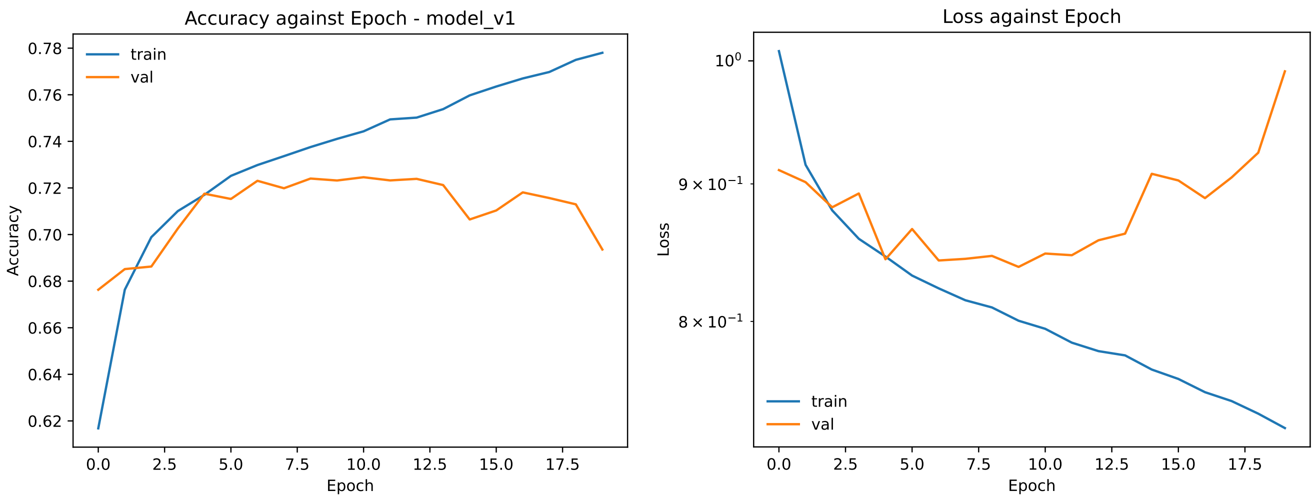

Figure 1: Left Plot: Accuracy against epoch curves for training and validation accuracy. Right plot: Loss against epoch curves for training and validation loss. Both plots show clear signs of overfitting by the trained CNN model.

The model achieves a final validation accuracy of 73.52% and a validation loss of 0.82. From epoch 3 onwards, training accuracy continues to climb while validation accuracy starts to plateau at around 72% accuracy, indicating mild overfitting. This divergence is consistent with the known difficulty of training CNNs on sparse jet images (de Oliveira et al. 2016). The fact that most of the image pixels are zero introduces redundant spatial information that makes it harder for convolutional filters to capture the discriminating substructure features (Qu & Gouskos 2020).

4.2 Confusion Matrix

The normalised confusion matrix evaluated on the test set is shown in Figure 3. The diagonal entries represent the fraction of jets that have been correctly identified, whilst the off-diagonal entries represent misclassification rates.

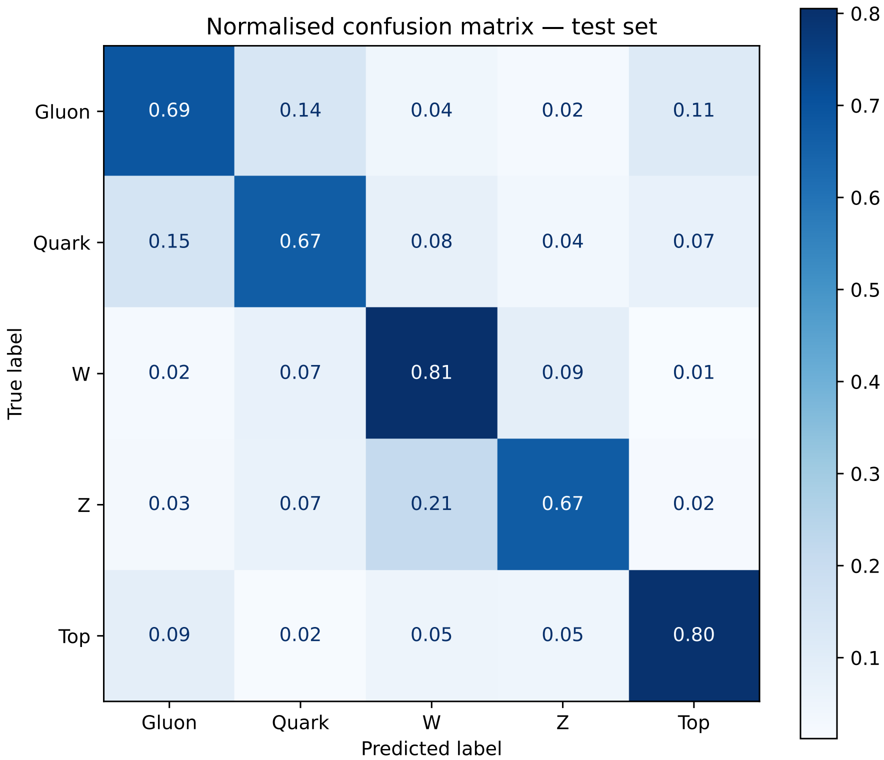

Figure 2: Confusion matrix for the trained CNN model. The model achieves recall values of 0.69 (Gluon), 0.67 (Quark), 0.81 (W), 0.67 (Z), and 0.80 (Top)

The largest off-diagonal entry is Z-W misclassification which is at 21%. This very high misclassification rate is physically expected as W and Z bosons have nearly identical two-prong hadronic decay structures with masses differing by 11GeV - 80.4GeV and 91.2GeV respectively (Workman, R.L. et al. 2022). Due to this similarity, this makes these two bosons the hardest pair to separate from calorimeter pixel patterns alone. This is consistent with Thaler, J et al. (2011) who treat W and Z jets as effectively the same category in N-subjetiness studies due to their kinematic similarity. The next largest source of confusion is the Gluon-Quark misclassification at 14% which reflects the known difficult of quark-gluon discrimination as a result of their similar jet substructure, namely transverse momenta (Gallicchio, J. and Schwartz, M.D. 2013).

4.3 ROC curves and AUC

The receiver operating characteristic (ROC) curve plotting for each of the five classes was also conducted to see how well the model separates the classes across all possible probability thresholds as a standard evaluation framework for jet image classifiers (de Oliveira et al. 2016).

The ROC curves for all five classes are shown in Figure 4, with both linear and logarithmic false positive rate axes.

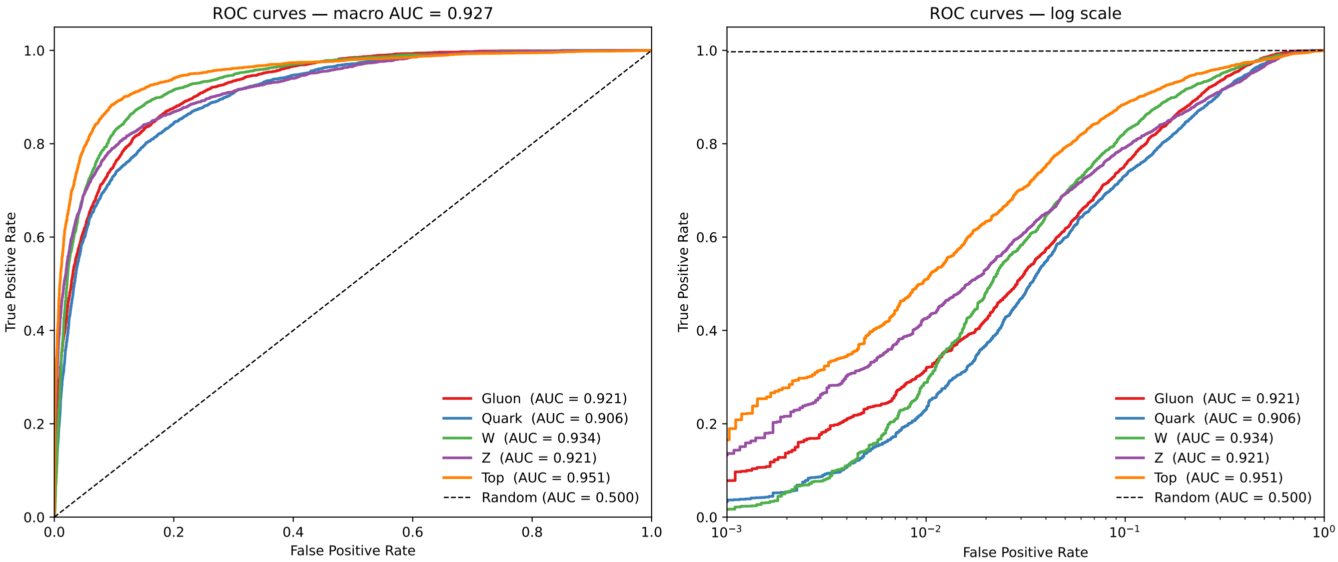

Figure 3: ROC curves for all five classes: linear (left plot) and logarithmic (right plot) false positive rates. The macro-averaged AUC is 0.927. per class AUC values are: Gluon (0.921), Quark (0.906), W (0.934), Z (0.918) and Top (0.948)

The macro-average AUC of 0.927 greatly exceeds the random baseline of 0.5. This result is comparable to simple MLP baselines on the same dataset (Moreno, E.A. et al. 2020), which confirms that raw 100x100 calorimeter images carry discriminating information for jet classification as initially proposed by de Oliveira et al. (2016). Top quark jets achieve the highest AUC of 0.951, consistent with their three-prong structure (Workman, R.L. et al. 2022), which provide the most distinctive spatial pattern in the calorimeter image.

The log-scale ROC curves reveal that the class separation is more pronounced at physically realistic operating points of false positive rate (FPR) < 0.01. Top jets maintain meaningful true positive rates at FPR ≈ 0.001, whilst light quark jets show the steepest degradation at low FPR, confirming that light quark jet identification is most challenging at tight selection requirements.

4.4 Efficiency as a function of jet feature variables

To further investigate the CNN's classification performance, the model is tested against jet feature substructure observables by identifying per-class efficiency, defined as the fraction of jets of each true class correctly identified, as a function of three jet feature variables: jet mass

m_{jet}, transverse momentum p_T and constituent multiplicity. The results are shown in Figures 4 and 5. The error bars are derived from binomial statistical uncertainty given by (Barlow, R. 1989):

$$\sigma_\epsilon = \sqrt{\frac{\epsilon(1-\epsilon)}{n_{total}}} \tag{3}$$

where ϵ is the efficiency and n is the number of jets in the bin.

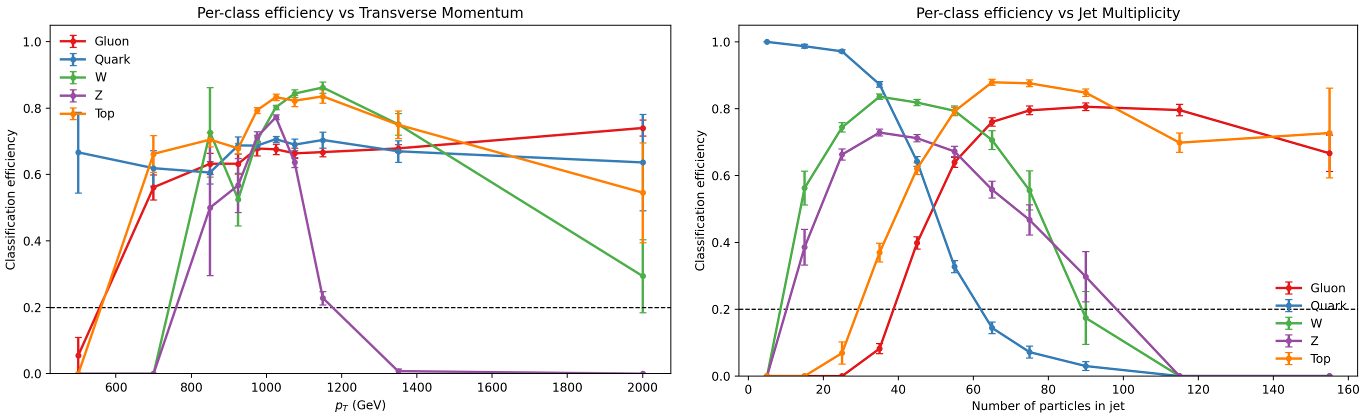

Figure 4: Plot of per-class efficiency vs Transverse Momentum (left plot) and per-class efficiency vs jet multiplicity (right plot)

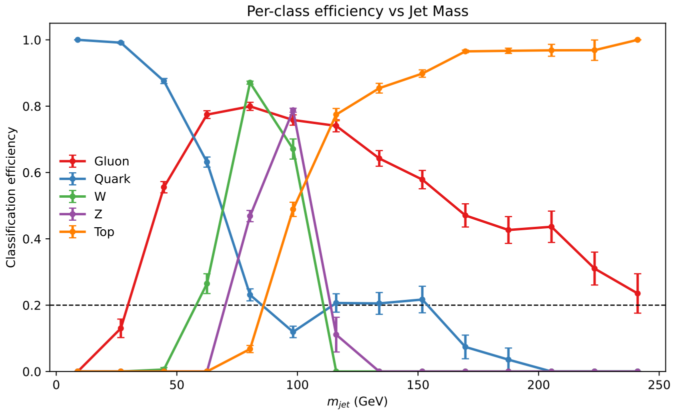

Figure 5: Plot of per-class efficiency vs jet mass

4.4.1 Efficiency versus transverse momentum

The efficiency as a function of transverse momentum is shown on the left plot of Figure 4. W and Z jet efficiency show a pronounced dependence on p_T, where both classes peak in efficiency at around 1000-1150 GeV and collapse sharply at high p_T with Z jet efficiency falling to zero above 1300 GeV, suggesting that the CNN model confuses Z bosons with W bosons in the boosted regime, reflected in the 21% misclassification rate described in this model's confusion matrix. This observation confirms the known challenge of W/Z distinction in high-boost regimes (Thaler & Van Tilburg. 2011, Larkoski et al. 2020).

4.4.2 Efficiency versus constituent multiplicity

The efficiency as a function of constituent multiplicity is shown on the right plot of Figure 4, in which the results show clear class separation consistent with known decay topologies of each particle.

Light quarks dominate at low multiplicity (under 30 particles) then collapses with increasing multiplicity. This observation is consistent with their narrow, sparse shower structure arising from the smaller QCD colour factor $C_F = 4/3$ compared to gluons ($C_A = 3$) which causes quarks to radiate fewer secondary particles (Gallicchio, J. et al. 2013). W and Z jet efficiency peaks at 25-40 constituents, as both decay to two quarks, and both drop sharply above 50, which reflects their two-prong structure (Larkoski et al. 2020). Top quark jets efficiency rises and peaks around 60-90 constituents, which reflect their three-prong topology (Kaplan, D.E. et al. 2008). Gluon jet efficiency is broad across 40-120 constituents, which agrees with wider and more diffuse gluon-initiated shower producing many soft particles (Gallicchio, J. et al. 2013).

4.4.3 Efficiency versus jet mass

The efficiency as a function of jet mass is shown in Figure 5. Light Quark jet efficiency peaks in the range 0-30GeV, consistent with light quarks producing low-mass jets. W boson jets efficiency peaks at approximately 80GeV and Z boson jet efficiency peaks at approximately 91GeV, which is in close agreement with known values of 80.377GeV and 91.1876GeV (Workman, R.L. et al. (Particle Data Group) 2022). Top quark jet efficiency rises continuously above 150 GeV, consistent with top quark mass of approximately 173 GeV producing high-mass jets.

Thus, this result demonstrates that the CNN has implicitly learned characteristic mass scales of each particle just from raw calorimeter deposits alone, which is consistent with the findings of de Oliveira et al. (2016), who demonstrates that CNNs trained on calorimeter images can learn meaningful substructure features, directly from spatial energy patterns.

However, there is a clear limitation presented in the distinction of transverse momenta in highly boosted W and Z bosons, which substantiates the Z-W confusion. As a result, this motivates the features DNN approach, where explicit access to jet feature observables provide more discriminating information that cannot be reliable inferred from merged pixel-level deposits alone.

5 Results - Features DNN vs Image CNN

5.1 DNN - Training performance

Training and validation accuracy and loss curves are shown in Figure 6, with early stopping converging at 28 epochs. Both the training accuracy and loss curves are systematically lower and higher, respectively. This a known and expected consequence of dropout being active during training but not during validation inference (Srivastava et al. 2014).

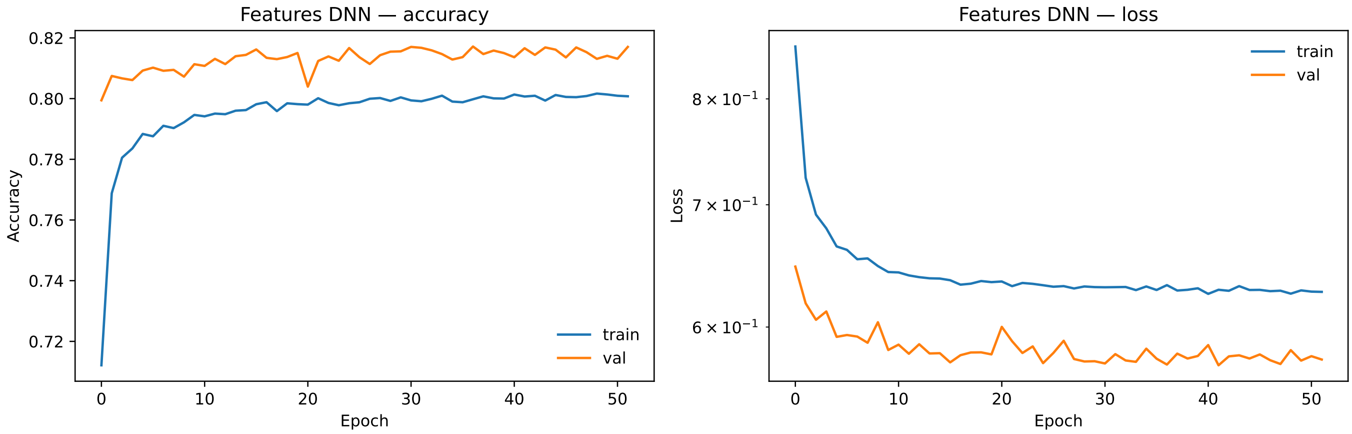

Figure 6

The model performance is therefore the validation of accuracy of 81% which represents a 9 percentage point improvement from the CNN image validation accuracy of 72%. The loss curves both show convergence with only random oscillations which indicates that the model is generalising well, with a rapid initial convergence within the first 5 epochs. This is indicative of the DNN having access to physically motivated discriminants such as jet mass and N-subjettiness ratios, which are specifically designed to capture discriminating properties of each jet class, rather than inferring these properties from spatial pixel patterns as the CNN does.

5.2 Comparing Confusion Matrices

When comparing the confusion matrices that both the CNN and DNN produce, we see a very clear improvement in performance from the features DNN across all true classification rates (see Figure 7) and modest improvements in most of the misclassification rates, with the most striking improvement being Z boson jet identification.

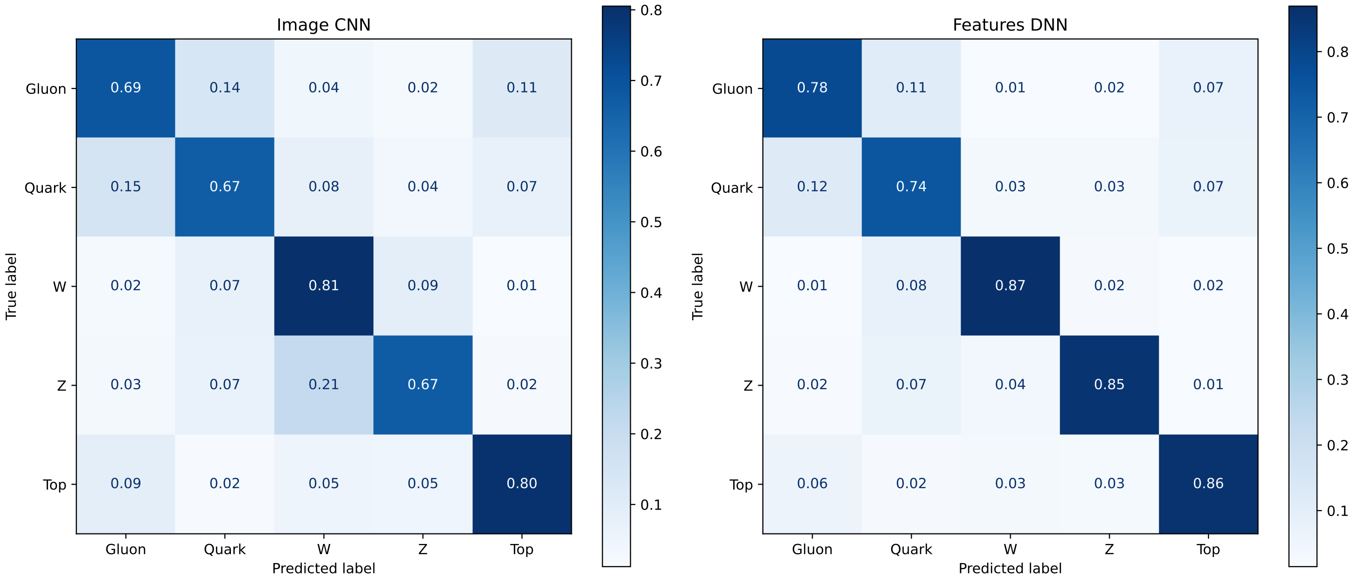

Figure 7: Confusion matrices for both CNN and DNN. For the DNN, per-class recall increases from 0.69 to 0.78 (Gluon), 0.67 to 0.74 (Quark), 0.81 to 0.87 (W), 0.67 yo 0.85 (Z), and 0.80 to 0.86 (Top), demonstrating an overall improvement in misclassification rates and thus performance from the DNN as compared to the CNN.

Z-W misclassification drops from 21% in the image CNN to 4% in the features DNN, which is a fivefold reduction and the most significant change in the entire confusion matrix. This result further confirms the observations made by de Oliveira et al. (2016), that raw pixel representation classification through CNN's is insufficient to fully resolve subtle differences between kinematically similar particles. Since the DNN has access to more discriminating capacity in groomed jet mass variables, it is able to resolve the Z-W confusion present in the CNN, which itself appears most prominently when discriminating Z- and W- bosons based on their transverse momenta.

5.3 Comparing ROC curves and AUC

The per-class ROC curves further illustrate the greater discriminating power in the DNN model compared to the CNN model. The features DNN achieves a macro-averaged AUC of 0.962 (see Figure 8), which is an improvement of 0.035 from the attained value of 0.9227 from the CNN.

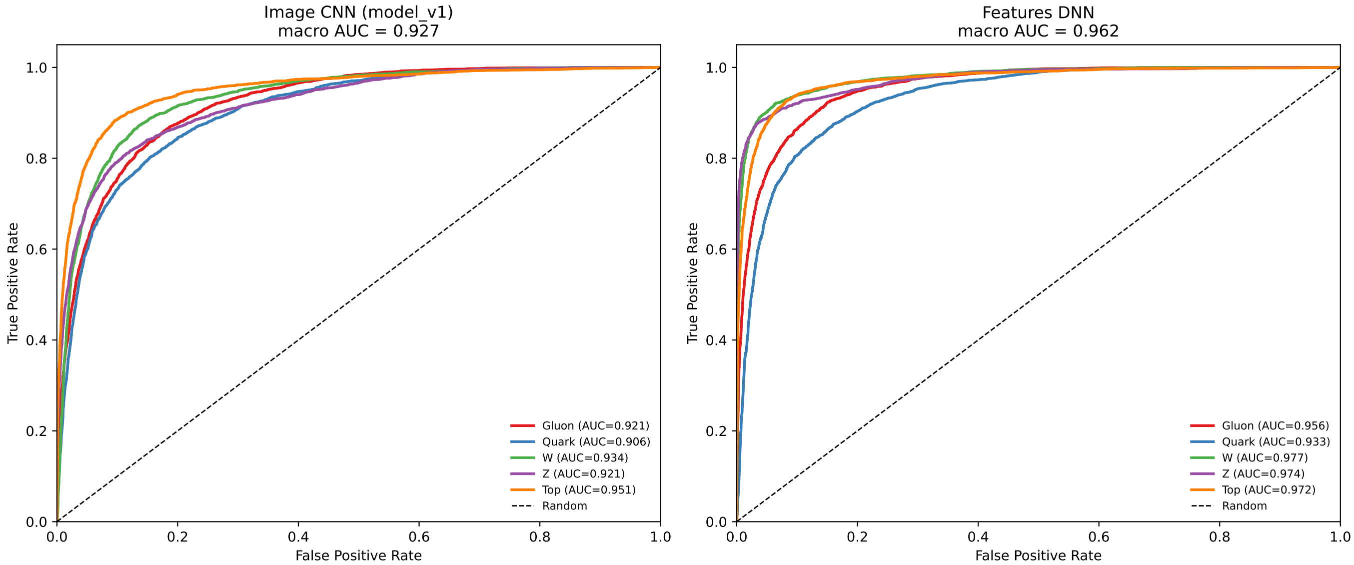

Figure 8: ROC curves comparison between CNN and DNN. The macro AUC for the DNN (0.962) is noticeably higher than the macro AUC for the image CNN (0.927) suggesting an overall better performance in true classification rate in the DNN versus the CNN model.

Across each of the five classes, the DNN exhibits improvement in true positive classification rates (as summarised in Table 3), demonstrating the DNN trained on jet feature observables as a stronger classifier than a CNN trained on raw calorimeter images.

Table 3: Per-class AUC comparison between image CNN and features DNN

Class	**Image CNN**	**Features DNN**	**Gain**
Gluon	0.921	0.956	+0.035
Quark	0.906	0.933	+0.027
W	0.934	0.977	+0.043
Z	0.918	0.974	+0.053
Top	0.948	0.972	+0.021
Macro	**0.927**	**0.962**	**+0.035**

6 Discussions

6.1 Conclusion

Overall, this study demonstrates that a CNN, trained only on raw 100x100 calorimeter images, not only can discriminate between kinematically similar particles effectively but imply jet feature substructure without any foreknowledge. The image CNN achieves 72% accuracy and macro-averaged AUC of 0.927, demonstrating that calorimeter spatial structure alone is sufficient for jet classification. Despite these positive results, the DNN, trained only on 53 jet substructure features, maintains a stronger performance with an 81% validation accuracy and macro-averaged AUC of 0.962. The CNN struggles to separate W and Z jets at high transverse momentum, reflected in the 21% Z-W misclassification rate arising from in an inability for pixel-level representations to resolve the ≈11GeV W/Z mass difference at high boost. In contrast, the features DNN yields a five-fold reduction in Z-W misclassification from 21% to 4%, which is directly attributed to the explicit jet mass observables, that the DNN is trained on.

These results therefore demonstrate that while raw calorimeter images contain rich physically interpretable spatial information, explicit jet substructure observables still provide substantially greater discriminating power for kinematically similar particles.

6.2 Experimental Improvements

It is worth noting that that the effective sample size in this study was significantly limited by the system RAM of the operating device used. This prevented any more than the 30,000 validation and 80,0000 training and test samples to be used for either the CNN or DNN. In comparison, de Oliveiria et al. (2016) trained their CNN on approximately 500,000 jets for their main result, and noted that performance improved with dataset size. In addition, datasize requirements scale with model complexity and input dimensionality (LeCun et al. 2015). Since the 4.8 million parameter CNN has a high parameter-to-sample ratio, it is not surprising that the model elicits mild overfitting from epoch 3 onwards. As a result, running both of these models on a device with higher RAM and GPU memory will allow for more reliable and robust results, due to the significant increase in input sample size.

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